Green’s functions of degenerate Hubbard ground states using Liouvillian recursion

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Abstract

An implementation of Liouvillian recursion for many-body Green’s functions is extended to the degenerate ground-state ensemble of small odd-site Hubbard chains. The required ensemble averages are obtained from separate pure-state calculations. A fermionic mode permutation relates the two spin partners, allowing the second preparation circuit to be constructed from the first without directly preparing a mixed state. The implementation combines Jordan-Wigner operator mappings, variational state-preparation circuits, Pauli-observable estimation, and classical reconstruction of Green’s functions. Three- and five-site calculations are compared with exact diagonalization; three-site hardware results were obtained on IBM Quebec. The reported calculations establish a working extension of the existing workflow for these systems. Finite recursion depth, approximate state preparation, and sampling or hardware errors remain distinct limitations.

1. Model and target ensemble

Consider the open-boundary, spin-independent Hubbard Hamiltonian, with the hopping amplitude defining the energy unit,

$$H = -t \sum_{\langle i,j\rangle,\sigma} (c^{\dagger}_{i\sigma} c_{j\sigma} + c^{\dagger}_{j\sigma} c_{i\sigma}) + U \sum_i n_{i\uparrow} n_{i\downarrow} - \mu \sum_{i,\sigma} n_{i\sigma} \quad (1)$$

The three-site implementation uses $t = 1$, $U = 4$, and $\mu = U/2 = 2$. Fermionic modes are ordered in two spin blocks, $(0\uparrow, \ldots, (N-1)\uparrow, 0\downarrow, \ldots, (N-1)\downarrow)$, and mapped to qubits with the Jordan-Wigner transformation. An N-site chain therefore occupies $2N$ qubits: six for three sites and ten for five sites. The fermionic Hamiltonian and annihilation operators are constructed with Qiskit Nature and represented as sparse Pauli sums [2].

For the two-dimensional ground-state manifold considered here, the target is the equal-weight ensemble of orthonormal spin partners. This is a specified zero-temperature ensemble, rather than a claim that every degeneracy must be averaged with these weights:

$$\rho_0 = \tfrac{1}{2} \sum_{s=1,2} |g_s\rangle\langle g_s|, \quad \mathrm{Tr}(\rho_0 O) = \tfrac{1}{2} \sum_{s=1,2} \langle g_s|O|g_s\rangle \quad (2)$$

Equation (2) reduces each linear ensemble observable to two pure-state expectation values. A coherent superposition of the two states is generally not an equivalent replacement, because it introduces off-diagonal contributions in this basis.

2. Construction of the spin-related state

Let P exchange the two fermionic spin blocks. For a spin-independent hopping matrix and on-site interaction, this mode permutation preserves the Hamiltonian, $P^\dagger H P = H$. In the ground-state sector under investigation, the second spin partner is obtained as $|g_2\rangle = P|g_1\rangle$. The implementation realizes P as a sequence of adjacent fermionic swaps [3], with two-mode action

$$F|00\rangle = |00\rangle, \quad F|01\rangle = |10\rangle, \quad F|10\rangle = |01\rangle, \quad F|11\rangle = -|11\rangle \quad (3)$$

The minus sign on double occupation is essential for the induced transformation of fermionic modes. For three sites the permutation is [3, 4, 5, 0, 1, 2]. The routine successively moves each desired mode into place by adjacent swaps; the resulting block exchange requires N^2 adjacent FSWAPs before transpilation. This count is a property of this construction, not a hardware gate-count bound.

Starting from a fixed variational circuit V, the two preparation circuits are V and PV. The three-site script uses a six-qubit real-amplitudes ansatz with eleven repetitions and linear entanglement, with stored optimized parameters. Validation in the original internship report checks state overlap, energy equality, and the symmetry transformation. The reported overlap between the approximate spin partners is about 2×10^{-6}. This is a numerical diagnostic for those preparations, not a certified error estimate for their Green's functions.

3. Liouvillian recursion and ensemble reconstruction

The retarded single-particle Green's function is defined by

$$G^R_{ij,\rho}(\tau) = -i\Theta(\tau)\,\mathrm{Tr}[\rho\{c_i(\tau), c_j^\dagger\}], \quad c_i(\tau) = e^{iH\tau} c_i e^{-iH\tau} \quad (4)$$

The underlying hybrid algorithm [1] constructs a Krylov basis in operator space, with a state-dependent anticommutator product. To state the conventional frequency-domain recurrence, define $K(A) = [A,H]$; the code uses $L(A) = [H,A] = -K(A)$, so recurrence signs must be kept consistent with that convention. With $\langle A,B\rangle_\rho = \mathrm{Tr}[\rho\{A^\dagger,B\}]$, normalized operators obey the stationary-state Lanczos relation

$$R_{n+1} = KQ_n - a_n Q_n - b_n Q_{n-1}$$
$$a_n = \langle Q_n, KQ_n\rangle_\rho, \quad b_{n+1} = \sqrt{\langle R_{n+1}, R_{n+1}\rangle_\rho} \quad (5)$$

Here $Q_{-1} = 0$, $b_0 = 0$, and $Q_{n+1} = R_{n+1}/b_{n+1}$ until termination. The code's inner-product interface writes the conjugate convention $\mathrm{Tr}[\rho\{A,B^\dagger\}]$. For a stationary density operator, the Liouvillian admits the corresponding Hermitian tridiagonal representation on the non-null Krylov subspace. An approximate preparation need not be exactly stationary; the ideal Lanczos properties should not be assumed to hold without numerical checks.

Diagonal resolvent elements are reconstructed as continued fractions from the tridiagonal coefficients, with $z = \omega + i\eta$ and $\eta > 0$. The hybrid polynomial reconstruction also provides off-diagonal matrix elements [1]. Each spin partner has its own state-dependent recursion. The reconstructed Green's functions are then combined as $G_{\rho_0} = (G_{g_1} + G_{g_2})/2$. Averaging the Lanczos coefficients before reconstruction would generally be incorrect, since the continued fraction is nonlinear in those coefficients.

4. Implementation and numerical observations

The *asaad* branch of Liouville-Lanczos contains the extension [2]. The three- and five-site scripts iterate over spin sectors, prepared ground states, and selected initial operators. Exact-diagonalization runs supply a reference calculation; estimator-based runs use the same Pauli-operator representation with either a simulator or the IBM runtime estimator. The three-site script selects the ground-state eigenspace with an energy tolerance of 10^{-9} and also uses a 10^{-9} simplification threshold in the Pauli workflow. These implementation tolerances are not accuracy guarantees for the reconstructed spectrum.

The contribution consists of integrating the degenerate ensemble, implementing the adjacent-FSWAP mode permutation and associated checks, extending the calculations to odd-site systems, and analyzing Green's functions and Galitskii-Migdal energy estimates. The Liouvillian-recursion method and the original software architecture are those of the AlgoLabs collaboration [1,2].

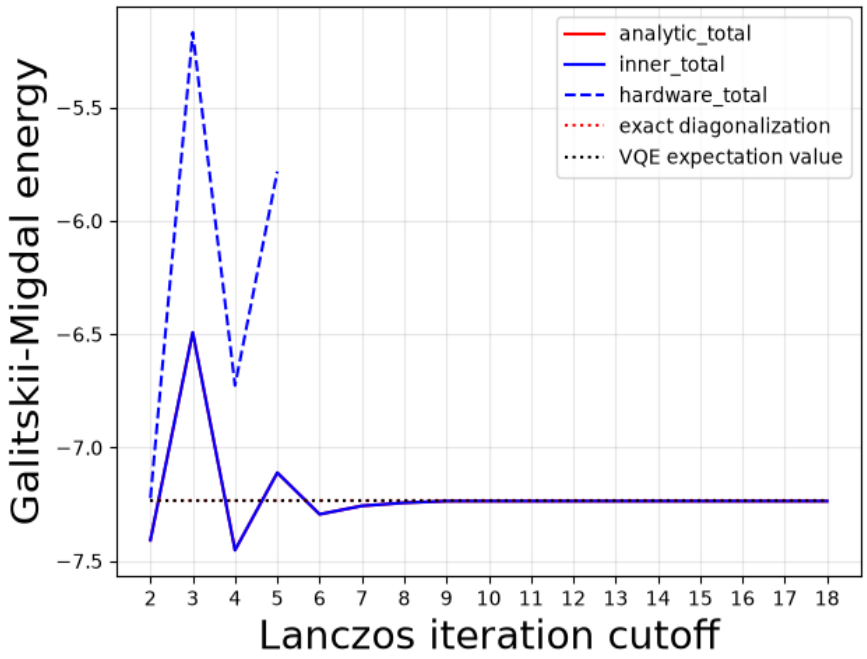


Figure 1. Three-site Galitskii-Migdal energy estimates versus Lanczos iteration cutoff, reproduced from the internship presentation assets. "analytic_total" denotes the matrix-reference recursion and "inner_total" the estimator-based calculation; horizontal references are the exact-diagonalization energy and the original variational expectation value. The hardware series is shown only at the available early cutoffs. The plot is retained as a historical numerical result; no new runs or uncertainty analysis were performed for this report.

The July report records three-site Green's-function convergence at approximately ten recursion steps. In contrast, the five-site Green's-function calculations became impractical at roughly seven to eight steps, before full convergence was obtained. An energy estimate and the full frequency-dependent Green's function need not converge at the same cutoff. Figure 1 illustrates the energy diagnostic; it should not be read as a uniform spectral-error bound or evidence that the hardware calculation converged at every cutoff shown for the classical curves.

The original data products include simulator and IBM Quebec runs. This report does not assign confidence intervals, shot counts, a fixed broadening η, or a hardware error budget where these have not been recovered from the archived run metadata. Quantitative noise robustness or computational advantage is therefore not inferred from the plotted agreement.

5. Symmetry reduction investigated during the internship

A separate investigation implemented the symplectic representation of Pauli symmetries and the associated Clifford transformations used in qubit tapering [4,5]. In that approach, commuting Pauli symmetries are identified from a binary symplectic kernel, independent commuting generators are selected, and symmetry-sector eigenvalues are fixed after a suitable Clifford basis change. This can reduce qubit requirements within a chosen symmetry sector.

This investigation did not supply the final ensemble treatment. Fixing a symmetry sector and evaluating a statistical mixture of degenerate states are different operations. In particular, a continuous spin-rotation symmetry should not be identified with a set of independently taperable Pauli generators. Conversely, the presence of spin symmetry does not rule out useful discrete Pauli symmetries; the applicable generators and sectors must be established for the mapped Hamiltonian. The workflow used here instead obtains the required spin partners explicitly and combines their observables.

6. Conclusions

For the odd-site Hubbard systems studied, the ensemble Green's function can be evaluated without a mixed-state preparation circuit by exploiting a known spin-related ground-state pair. The fermionic permutation supplies the second circuit, and independent state-conditioned recursions supply the two Green's functions. The three-site implementation is supported by exact-diagonalization, simulator, and hardware comparisons. The five-site calculation exposes the growth of the classical operator-space workload before full spectral convergence. The construction is specific to an identified ground-state manifold and its known symmetry relation; it does not remove the general cost of ground-state preparation or establish quantum advantage.

References and implementation

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[2] AlgoLabs collaboration, Liouville-Lanczos, *asaad* branch. Three-site implementation: tests/cluster/3_site.py; permutation checks: tests/fswap.ipynb. The July internship report identifies commit 405ad22ebeb63e25e04d0508601615759092a8d2.

[3] F. Verstraete, J. I. Cirac, and J. I. Latorre, "Quantum circuits for strongly correlated quantum systems," Physical Review A **79**, 032316 (2009). DOI: 10.1103/PhysRevA.79.032316.

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[6] A. Hanna, *Computing Green's functions for degenerate ground state systems using Liouvillian Recursion*, AlgoLabs summer internship report, July 29, 2026 (unpublished). Numerical observations and Figure 1 in the present technical report are drawn from that internship and its presentation assets.